



TEL AVIV אוניברסיטת
UNIVERSITY תל אביב

Comparison and Evaluation of Visualizations

+

Story Telling

Visualization of Information

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Logistics

- Deliver the assignment by Sunday 19th.
- There will not be session from 14:00 to 16:00 on July 15th.

Instead:

- Two options for the presentation:
 - July 15th. from 18:00-20:00
 - July 16th. from 20:00 – 21:00

#	Content	Comment
1	High level overview of the steps of ML and discern when and why visualizations can be used to improve the process	Theory
2	Introduction: Why/What/How visualization pipe, semiology, variables, and taxonomy	Theory
3	Principles of human perception, principles of information processing and perception, saliency, etc.	Theory
4	Data sources and processes of data processing (pre-processing – data transformations)	Theory
5	Visualization methods for temporal data (time-series) and multi-dimensional data (multi variate)	Theory + Hands On
6	Interactive Visualizations with Shiny R (as an example) and other platforms (PowerBI) – how to use gen AI tools to generate visualizations	Theory + Hands On
7	In depth: Principles of interaction, methods of interaction in visualization	Theory + Hands On
8	Lie and truth in information presentation and Dashboards	Theory
9	Visualization of statistical information (e.g. variance, etc.) and visual analytics in machine learning	Theory + Hands On
10	Visualization of text and documents Visualization methods for trees, graphs, and networks	Theory + Hands On
11	Comparison and Evaluation of Visualizations + Effective Visualization + Story Telling	Theory + Hands On
12	Presentations	
13		



Agenda

1. Why Comparison and Evaluation of Visualizations?
2. Which visualization technique(s) should I use to solve my problem?
3. The visualization problem
4. Structures for Evaluating Visualizations
5. Effective Visualizations
6. Benchmarking Procedures
7. Story Telling
8. Pyramid Principle
9. Logistics

Which visualization technique(s) should I use to solve my problem?

- Which visualization technique?
 - Under what conditions their technique is better than existing techniques?
 - Whether a modification made to a particular method improves or makes worse some aspect of the technique or system.
- ➔ Try to identify some of the components and procedures necessary to assess and compare the effectiveness of visualization techniques.

Visualization problem

- List the types of tasks a user may wish to perform with the aid of visualization
- User characteristics, such as level of experience, that can affect the success of a visualization technique
- The characteristics of data
- Special attributes of visualizations themselves that need to be considered in the evaluation
- Benchmark Process

User Tasks

- Each of these tasks can be performed to varying degrees using visualization.
- Needs to have a clear vision of the degree of accuracy with which it needs to be accomplished.
 - Is it essential to correctly identify every occurrence of an object?
 - What rate of classification or ranking error is acceptable?
 - At what resolution is the location of an object needed?
 - How many falsely labeled correlations are acceptable?
 - For what data characteristics is it important that errors be minimized?

User Tasks

Identify the specific actions to be accomplish with the help of visualization

Focus on specific object: Identify and Locate

- Identify: to recognize an object based on the characteristics presented
- Locate: to establish the position of an object

Focus on groups, more than one object:

- Distinguish: to determine dissimilarity between objects
- Categorize: to classify objects into distinct types
- Cluster and segment: cluster is to group similar objects based on some relationship, segment is to separate dissimilar objects
- Rank: place a group of objects in an order
- Compare: to examine the similarities and differences between two or more objects where ordering is not possible
- Associate: to relate two or more objects, such as linking temperature and location in weather maps
- Correlate—to find a causal or reciprocal relationship between two or more objects

User Characteristics

Users can be classified in many ways. In this case we focus on knowledge and skills, or Familiarity with:

- Domain: How much expertise does the user have with the domain of the data being explored?
- Task: How much experience has the user had in performing the desired task?
- Data: Has the user examined this data previously and formed a reasonable mental model of its contents, or is this her first exposure to it?
- Visualization technique: Is this the user's first attempt to interpret the data using this kind of visualization?
- Visualization environment: Has the user employed this tool (or a similar one) in the past?

User Characteristics

- Each aspect can be treated as a continuum, and each can influence the assessment of a visualization.
- Ideally, we would want the range of characteristics of the participants to be as similar as possible to the intended audience for the technique.
- The results should be evaluated by grouping people with similar characteristics.

Data Characteristics

- Characteristics of the data being visualized can have a high influence on the effectiveness of the visualization technique.
- Ideally a large variety of data sets should be tested, and they should span the range of characteristics found in the domain for which the visualization is being designed.

Data Characteristics

Parameters to consider:

- Type: All combinations of types that can occur should be tested
- Size: Tests should cover the normal size range, and extreme values for the size.
- Dimensionality: Test a visualization with all possible subsets of dimensions (slices or projections), as well as, with the normal dimensionality.
- Number of parameters: Test with both univariate and multivariate data sets (unless only univariate data exists)
- Structure: can be simple (e.g. table) or complex (e.g. hierarchical). Test using all commonly encountered structures within the domain
- Range: Test the entire range of possible values, including the extremes of the range.
- Distribution: Data can be uniformly or nonuniformly distributed, both in values and in attributes (such as spatio-temporal position). Tests should involve data sets that include distributions usually encountered in the domain, as well as some extreme cases.
- Real vs. Synthetic. Many researchers perform evaluations using synthetic data, as it allows tight control of the data characteristics. In general, it is better to use real data that has the characteristics of typical data found in a particular domain.

Visualization Characteristics

- Computational performance
- Memory performance
- Data limitations: What are the upper and lower bounds for the size and complexity of the data that can be visualized with this technique?
- Degree of occlusion
- Degree of complexity: What is the normal learning curve for the technique?
- Degree of usability: How easy is it to perform the task?
- Degree of accuracy: How frequently is the user successful or unsuccessful in performing the desired task with this technique?

Structures for Evaluating Visualizations

- Usability tests: effective, efficient, engaging, error tolerant, and easy to learn. Observing users attempting to perform tasks.
- Expert reviews:
 - **Experts in visualizations:** carried out with a small number (three or fewer) of qualified reviewers. Checklist of desirable features for an effective visualization, against which the object of the evaluation is assessed (heuristic evaluation).
 - **Domain experts:** familiar with the types of data and tasks found within their domain, and try to identify the degree to which the system or technique could accommodate the data and support the tasks
 - **Field tests:** performed in the natural environment of the typical user and may last for weeks or months
 - **Case studies and use cases:** validate the effectiveness of their techniques by showing real (or sometimes contrived) examples of how their method can be used in solving a particular problem or performing a given task

Benchmarking Procedures

Benchmarks can be quantitative (resulting in a number) or qualitative (resulting in a relative judgement).

1. Formulate a hypothesis such as: this algorithm executes faster than that algorithm (not too general, such as “is better”). “System A allows novice users to more easily identify clusters in data sets containing 5– 10 dimensions and 1000 to 10000 data points than system B.”
2. Design the experiments: create tests that vary only a single attribute at a time
3. Execute the experiments:
 - the training of human subjects are critical;
 - each participant should get a similar amount of instruction on performing the experiments , including identifying the procedure they should follow , the expected format of their responses, and the pace at which they should proceed
 - Amount of time provided should be reasonably constrained, as users often will respond differently, given differing time constraints
 - The audience should be of sufficient size to make the results statistically significant
4. Analyze the results and validate the hypothesis

Designing Effective Visualizations

Goal: Communicate information **accurately, efficiently, and clearly**. Design should match the **purpose** of the visualization: Exploration

- Confirmation (analysis)
- Presentation (communication)
- Choose appropriate: **Visual encodings** (how data is mapped to graphical elements)

Interactive techniques (how users explore the data)

- There is **no single best visualization**—the best design depends on: Data characteristics
- User tasks
- Target audience

Steps in Designing Effective Visualizations

1. **Map data to visual attributes**

- Decide how data will be represented (position, color, size, shape, etc.).

2. **Design user interactions**

- Enable users to explore, filter, zoom, and modify the view.

3. **Select the right amount of data**

- Show enough information without overwhelming the user.

4. **Provide supporting information**

- Include titles, labels, legends, scales, and color keys to aid interpretation.

5. **Consider visual aesthetics**

- Create a clear, consistent, and visually appealing design.

6. **Evaluate the overall design**

- Ensure every component works together to communicate the intended message effectively.

Intuitive Mappings from Data to Visualization

1. Choose visual encodings that match:
 - **The meaning of the data**
 - **The user's domain knowledge and mental model**
2. Intuitive mappings: Reduce interpretation effort
 - Speed up understanding
 - Reduce the risk of misinterpretation
3. Consistency with user expectations makes visualizations easier to learn and use.

Examples of Intuitive Mappings

1. **Spatial data** → Screen position (e.g., maps using latitude and longitude).
2. **Time** → Animation or ordered sequence.
3. **Temperature** → Color (e.g., red = hot, blue/white = cold, depending on convention).
4. **Temperature or pressure** → Height or line length (similar to a thermometer).
5. Some mappings are **domain-specific**, where conventions already exist (e.g., cartography, geology, medicine).

Match Data Types to Visual Attributes

1. Select visual attributes that are compatible with the **data type**.
2. **Ordered (quantitative or ordinal) data** should use ordered visual attributes:
 - Position
 - Length
 - Size
 - Brightness
3. **Categorical (nominal) data** should use categorical attributes:
 - Shape
 - Color hue
 - Symbols
4. Avoid mismatches, such as:
 - Ordered data → Shape
 - Categories → Length

When to Use Non-Intuitive Mappings

1. Non-intuitive mappings can sometimes reveal **hidden patterns**.
 - Example: Mapping **time to color** can expose changes in speed or progression.
2. Best practice: Use **intuitive mappings as the default**.
3. Allow users to **customize mappings**, especially during exploratory analysis.
4. Balance **familiarity** with **flexibility** to support data discovery.

Selecting Effective Views

1. A **single view is rarely enough** for complex datasets. Select views based on:
 - **Data characteristics**
 - **User tasks**
 - **Target audience**
2. Provide the views users are most likely to need. Views should:
 - **Be clearly labeled**
 - **Be easy and quick to switch between**

Essential View Modifications

1. **Zoom and Pan:** Explore large or detailed datasets.
2. **Color Controls:** Change color palettes or customize colors.
3. **Mapping Controls:** View the same data using different visual encodings.
4. Different mappings can reveal **different patterns or insights.**

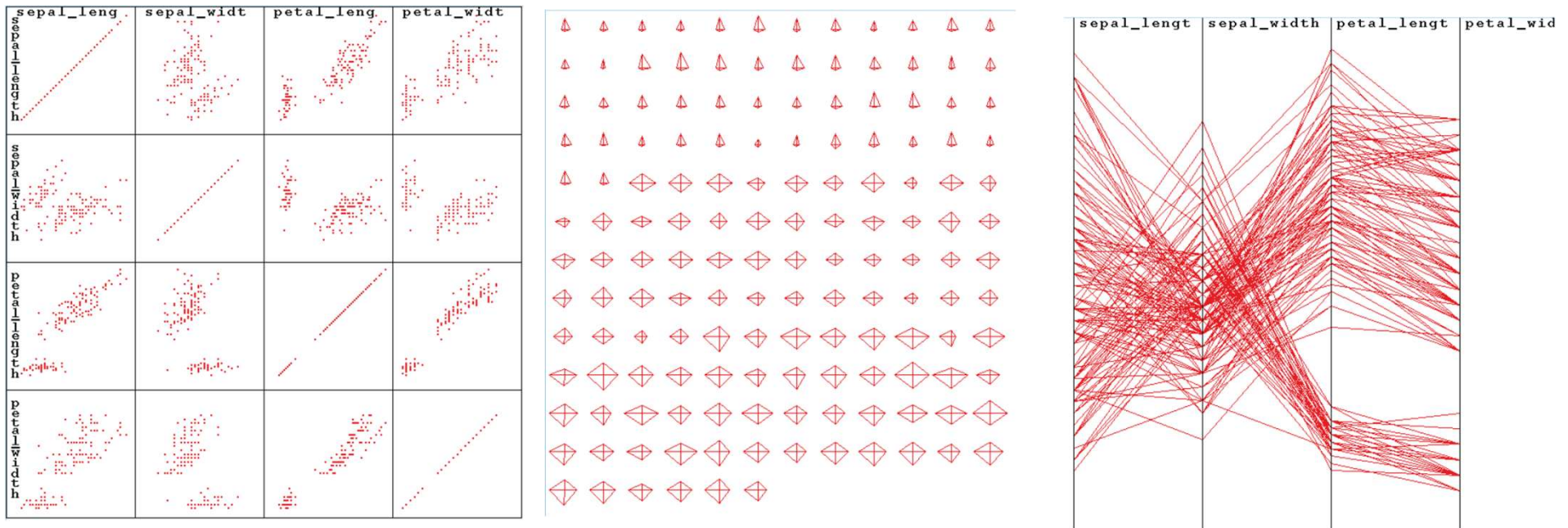
Supporting Data Exploration

1. **Scale Controls:** Adjust value ranges and distributions.
2. **Filtering and Clipping:** Focus on relevant subsets of the data.
3. **Level of Detail:** Show more or less information depending on the task.
4. Enable users to move easily between overview and detail.

Designing Easy-to-Use Interactions

1. View manipulations should be:
 - **Easy to learn**
 - **Easy to remember**
 - **Accurate and responsive**
2. Prefer **direct manipulation** whenever possible:
 - Interact directly with the visualization (e.g., drag to pan, scroll to zoom).
3. Keep interactions: **simple, intuitive, and consistent** across the interface.

Designing Easy-to-Use Interactions



Three views of the Iris data set (scatterplot matrix, star glyphs, and parallel coordinates).

(Image from XmdvTool.) – Ware Figure 13.2. , page 411

Finding the Right Information Density

A key design decision is how much information to display.

Two extremes:

- Too little information → unnecessary or gratuitous graphics.
- Too much information → cluttered, difficult-to-read visualizations.

Goal: Present enough information to support the task without overwhelming users.

Avoiding Too Little Information

- Do not create a visualization when simple text or a number communicates the message better.
- Avoid charts with only a few values.
- Avoid adding unnecessary derived values just to justify a graph.
- Choose the simplest representation that communicates the information effectively.

Avoiding Too Much Information

Excessive information can:

- Confuse or overwhelm users.
- Hide important patterns.
- Make it difficult to know where to focus attention.

Solutions:

- Allow users to show/hide information.
- Use multiple coordinated views instead of one overloaded display.

Managing Large and Dense Datasets

Reduce clutter by:

- Filtering irrelevant or less important data.
- Scaling data to improve distribution of visual elements.
- Showing details only when needed.

Objective: Maximize clarity, not simply display more data.

The Importance of Keys, Labels, and Legends

Visualizations should provide enough supporting information for accurate interpretation.

Include:

- Clear title or caption
- Axis labels with units
- Tick marks and gridlines (when needed)
- Symbol keys
- Color legends or color bars

Supporting information helps users interpret the visualization correctly.

The Importance of Keys, Labels, and Legends

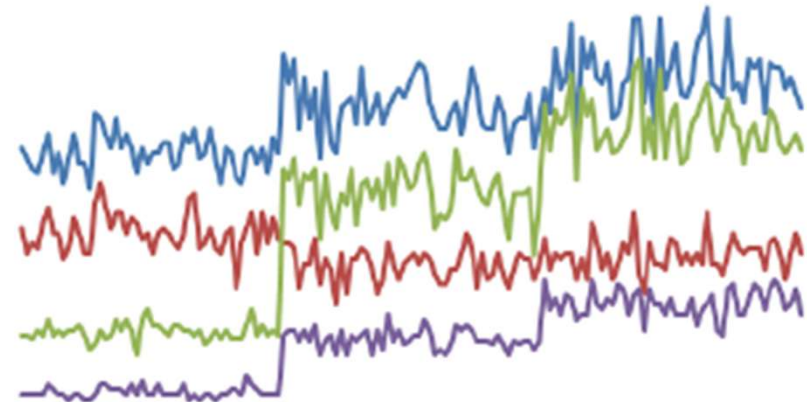
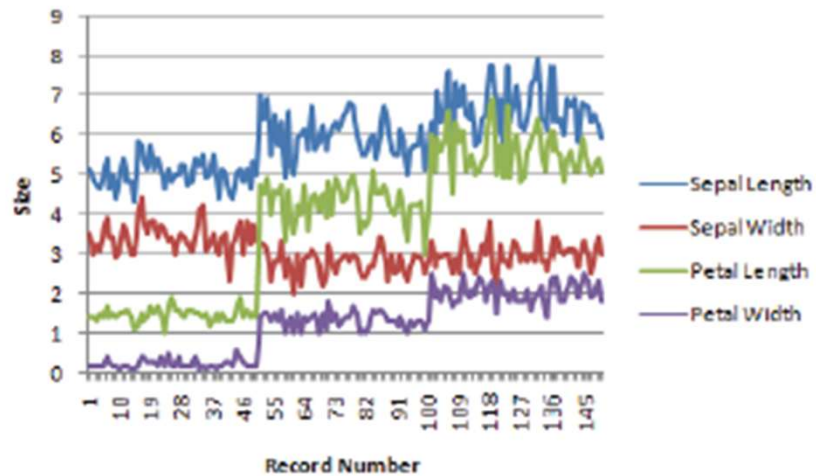


Figure 13.4. A complex visualization with and without captions/ticks/legends

Designing Effective Axes and Gridlines

Use gridlines and tick marks to improve value estimation.

Avoid extremes:

- Too many → clutter and occlusion.
- Too few → difficult to estimate values.

Place tick marks at meaningful intervals that match the data and user expectations.

Choosing Appropriate Value Ranges

Select axis ranges that:

- Represent the data accurately.
- Maximize readability and visual resolution.

Avoid misleading scales.

If the displayed range differs from user expectations (e.g., percentages not shown from 0–100%), clearly indicate the range.

Choosing Appropriate Value Ranges

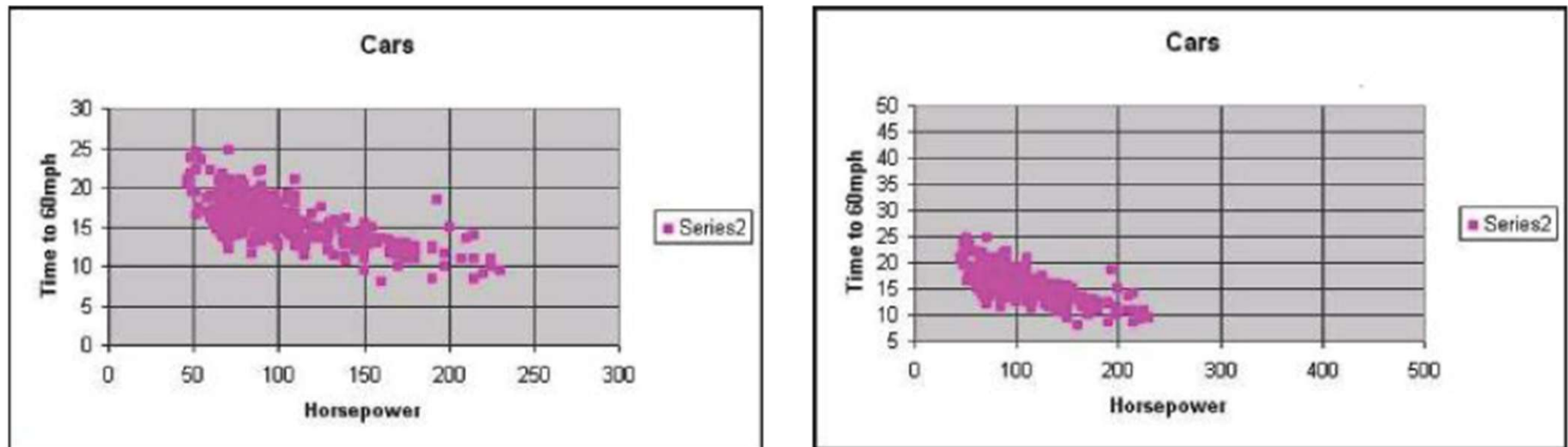


Figure 13.7. Logical and illogical data ranges.

Maintain Consistency Across Views

Use a consistent design across multiple visualizations:

- Same labels
- Same units
- Same grid style
- Same color meanings

If scales, ranges, or color mappings change, clearly communicate the changes.

Consistency reduces confusion and improves comparison between views.

Using Color with Care

Color is powerful but easily misused in visualizations.

Poor color choices can:

- Mislead users
- Hide patterns
- Reduce readability

Remember that color perception depends on context, and many users have color-vision deficiencies.

Guidelines for Effective Use of Color

- Keep the number of distinct color levels low for quantitative data.
- Use redundant encodings (e.g., color + size or color + position).
- Vary both hue and lightness for ordered color scales.
- Always provide a labeled color legend.

Choosing Meaningful Colors

Use colors that match users' expectations whenever possible.

Examples:

- Blue → Cold
- Red → Hot or Warning

Semantic colors are easier to learn, interpret, and remember.

Best Practices

- Design the visualization to work well even in grayscale.
- Add color only after the overall design is clear and effective.
- Test your visualization for readability and accessibility, including color-blind users.
- Use color to enhance communication, not to decorate.

Best Practices

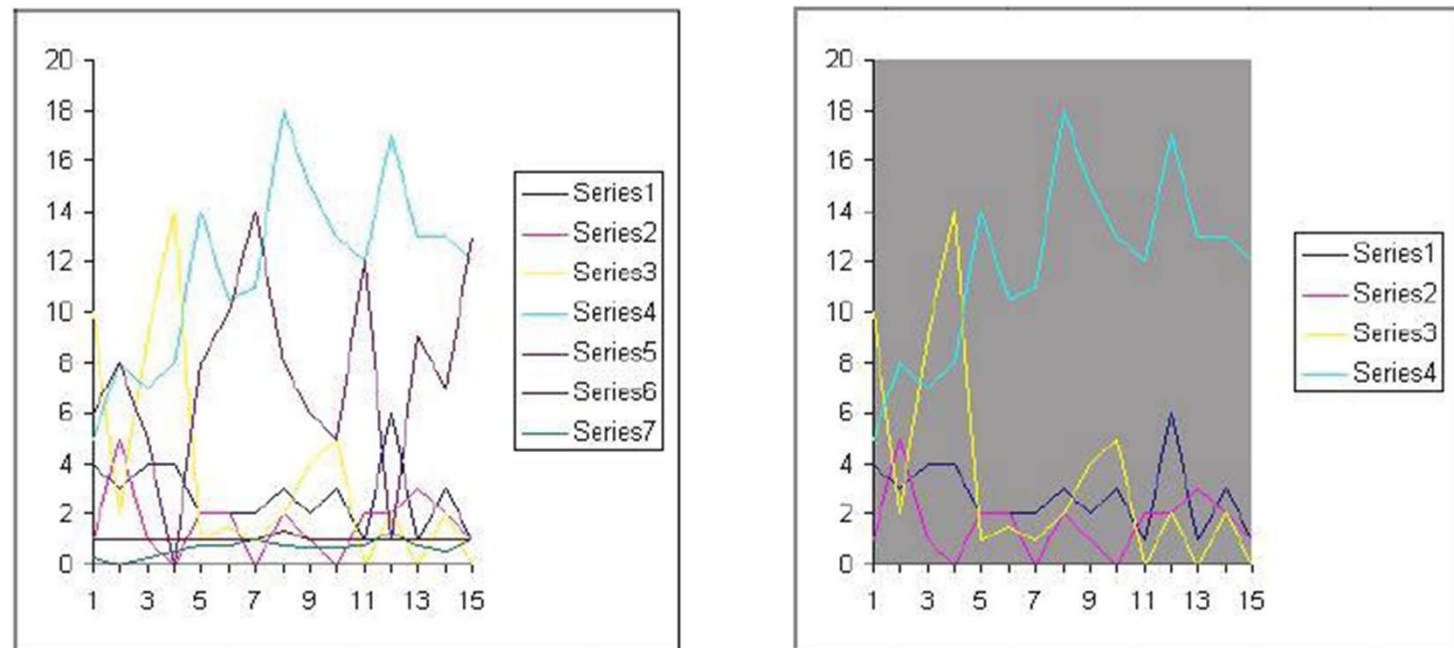


Figure 13.8. Too many colors versus a moderate number of colors

Remember Tufte 😊

Story Telling: How to build a compelling PPT & Data Visualization

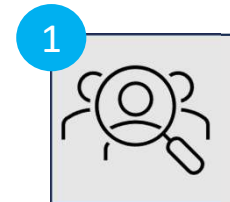
- Key Principle
 - Use a structured approach to guide the audience
 - Flow from high-level insights to detailed analysis
- Recommended Techniques
 - Storyboard Approach
 - Plan the flow of ideas slide by slide
 - Ensure logical progression and visual cohesion
 - Pyramid Principle
 - Start with the main message
 - Drill down into supporting arguments and data
- Why It Works
 - Builds a clear, persuasive narrative
 - Helps decision-makers quickly grasp key points
 - Enhances audience engagement and retention

What is the storyboard approach?

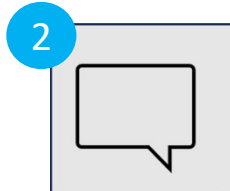
- Definition: A visual plan that outlines the flow and key points of a data-driven presentation
- Purpose
 - Provides a structured framework to
 - Organize data, insights, and narrative
 - Ensure clarity and audience engagement
- How It Works
 - Consists of a series of slides or panels
 - Each slide represents a major part of your analysis
 - Begins with the main message
 - Follows the Pyramid Principle: drill down into supporting details
- Why use it
 - Helps maintain a clear, logical progression
 - Enhances persuasiveness and comprehension

Storyboarding Order:

- Audience, goal & Settings: Define your audience, goal and setting.
- Main message & arguments: Determine the main message and define main arguments
- Evidence: Work-out the main arguments by supporting them with evidence (data, visualizations)
- Situation, Complication, question: Describe the situation, formulate the complication and define the central question.



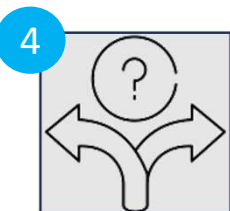
Audience, goal & Settings



Main message & arguments



Evidence



Situation, Complication,
question

Situation, Complication, question:

Situation:

Description of current situation. It should be:

- Recognizable
- Positioning
- Factual
- Not controversial
- Exciting

Complication:

A change in the current situation or a problem to solve

- Someone proposes a change
- An important development
- Expectations about the future

Question:

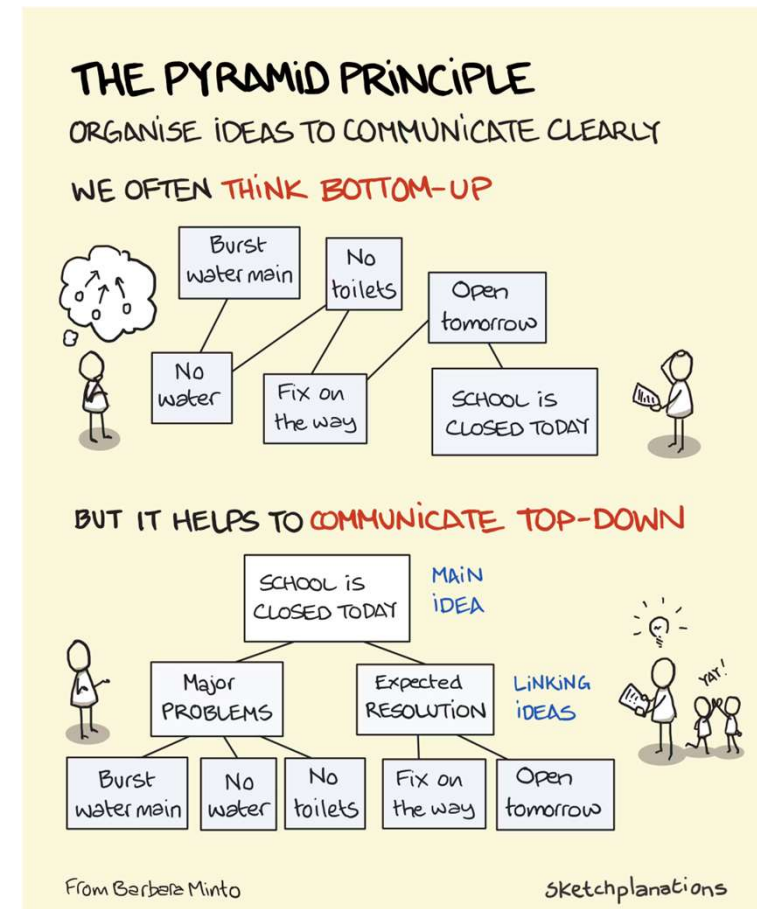
A translation of the complication in the central question.
Your presentation/document will give an answer on this question.

- Someone proposes a change
- An important development
- Expectations about the future

Story Telling using Visualizations

The Pyramid Principle

- What It Is
 - A logical framework for structuring complex ideas
 - Makes communication more clear, persuasive, and digestible
- Developed by Barbara Minto at McKinsey & Company
- How it works
 - Uses a top-down approach
 - Start with the main idea or recommendation
 - Follow with supporting arguments: Grouped logically in a hierarchical structure
- Why Use It
 - Ideal for business writing, analysis, and presentations
 - Helps build a strong, coherent storyline
 - Enhances clarity and impact for decision-makers



The Pyramid Principle - structure

- Start with the key takeaway or main message at the top.
- This is the answer to the main question or problem you're addressing.
- Imagine your audience only reads the first sentence. What's the one thing they should remember?
- Example: "Our analysis shows that we can increase customer retention by implementing a loyalty program focused on personalized rewards."

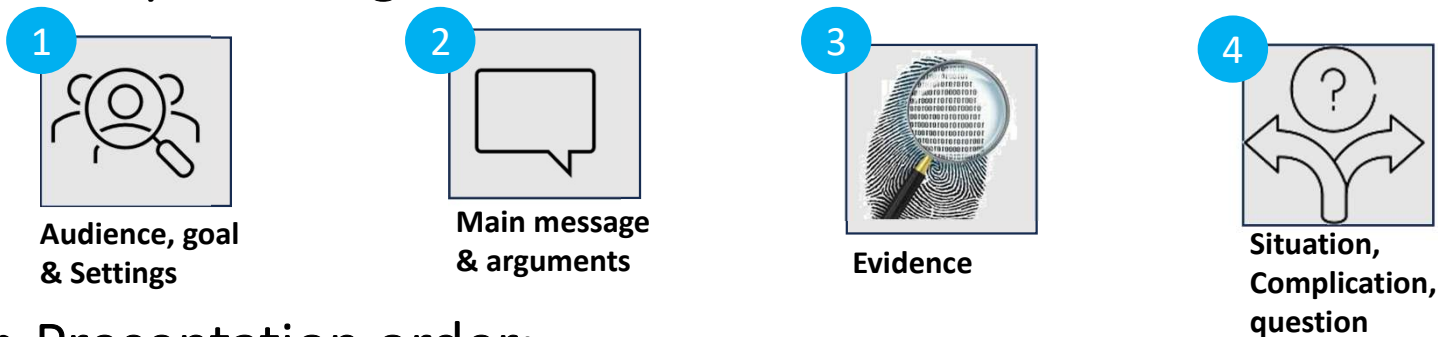
The Pyramid Principle - structure

- Begin with the Main Message (the “Answer”)
- Group Supporting Arguments (answer the “Why?” behind the main message)
- Mutually Exclusive, Collectively Exhaustive (MECE): Each argument should be distinct (mutually exclusive) and together cover all necessary points (collectively exhaustive).
- Supporting evidence with data and examples.
- Summarize and conclude

How to tell the story

The order in which you present the story (using the pyramid method) is different from the order in which you create the story.

- Storyboarding order:



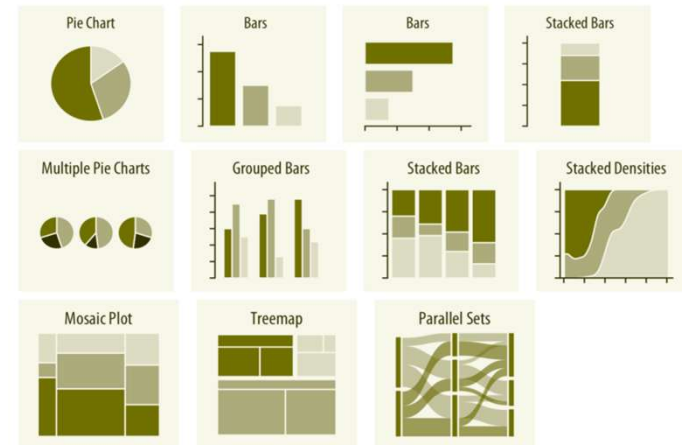
- Presentation order:



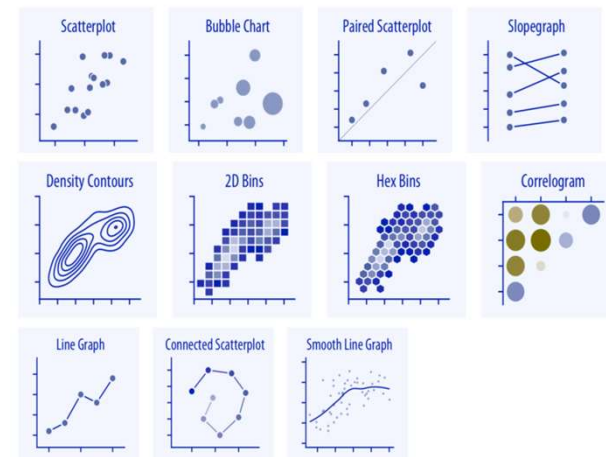
Visualization = tool to convey your message

1. Start with the message
2. Visualize what you can
3. Less is more
4. It is all about attention
5. Keep slides clean
6. Use the right charts & graphs

Proportions



Relationships

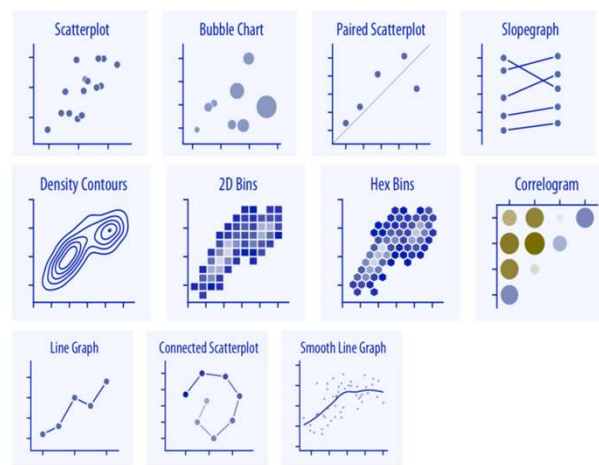


Use the right charts & graphs

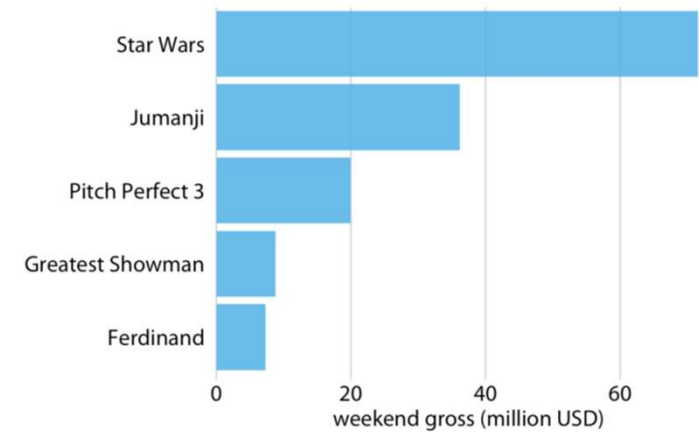
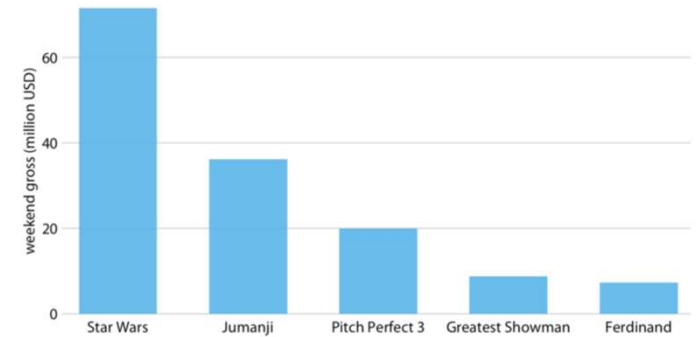
Proportions



Relationships

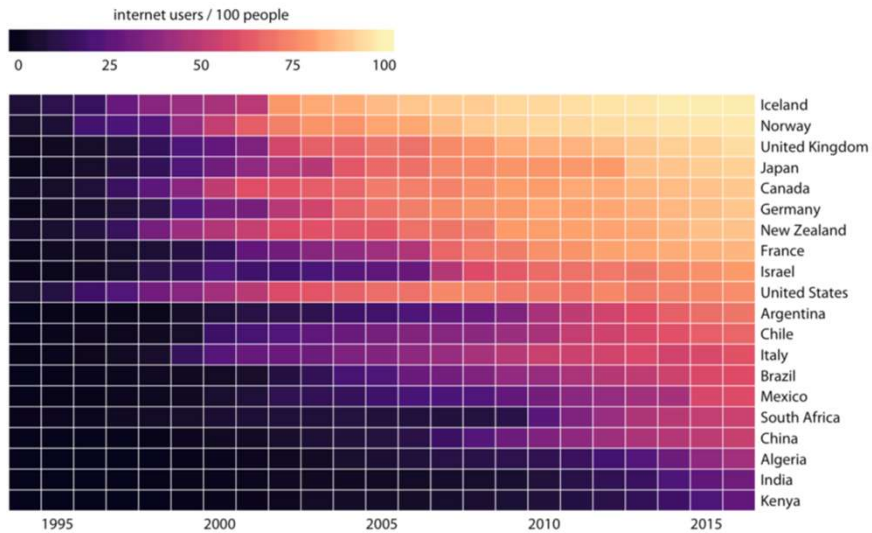


Amounts

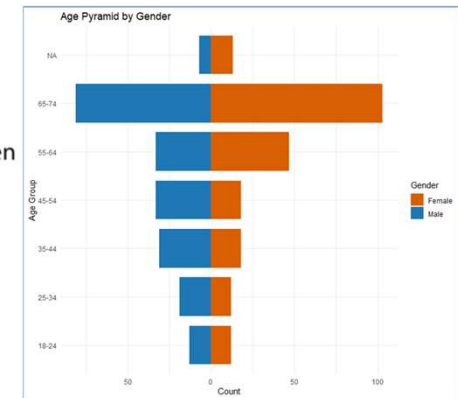
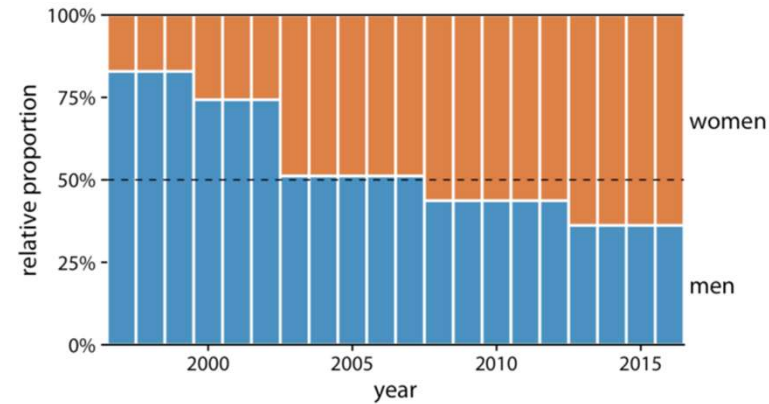


Use the right charts & graphs

Amounts

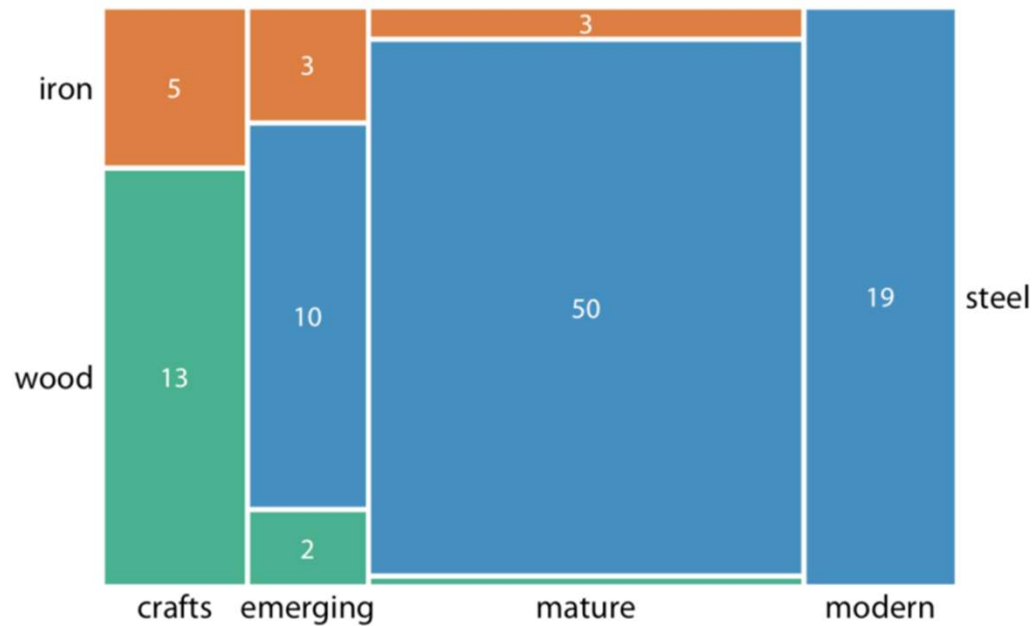


Proportions

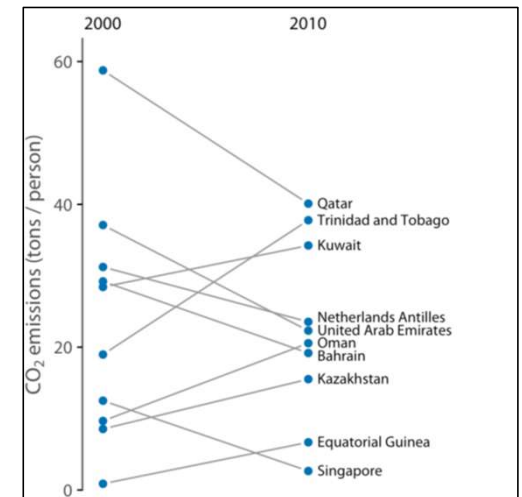
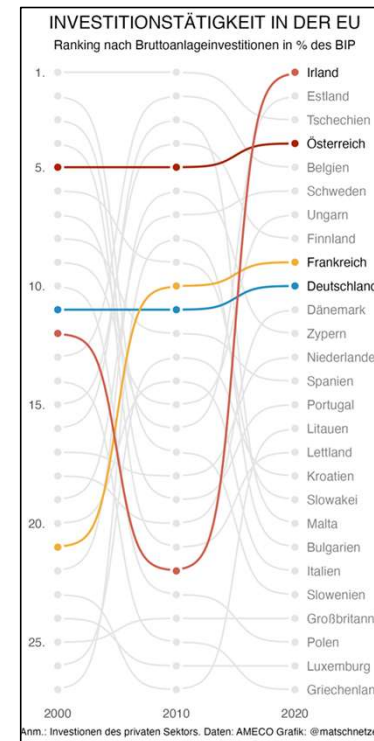


Use the right charts & graphs

Nested Proportions



Paired Data



Use the right charts & graphs

- Remember the visual Integrity – principles and violation principles

- Context
- Data-Ink

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